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REVIEW

Updates on Tau-Related Drug Targets and Potential Disease-Modifying Therapies for Alzheimer Disease

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Alzheimer disease (AD) is a chronic, multifactorial neurodegenerative disease affecting a large proportion of the elderly population with detrimental psychological and socioeconomic impacts. This review starts with a brief overview of the molecular aspects of AD. It aims to provide recent updates on several potential protein targets studied as valid therapeutic approaches such as Bassoon protein, triggering receptor expressed on myeloid cells 2 (TREM2), phosphorylated tau (chimeric molecules DEPTAC and PhosTAC), O-GlcNAcase (OGA), and misfolded protein seeds. This review emphasizes the importance of pursuing a multifaceted approach to diversify and synergize the current AD pharmacotherapy. *(Am J Pathol 2025, 195: 1975—1987; https://doi.org/10.1016/j.ajpath.2025.05.020)*

Tauopathies are a category of neurodegenerative diseases characterized by distinct brain lesions containing aggregates of filamentous abnormal microtubule-associated protein tau.[1] Diagnosing a tauopathy is based on a patient's clinical history, symptom onset, and progression.[2] Each tauopathy has distinct manifestations, but they share cognitive and motor impairment, behavioral and psychological alterations, visual symptoms, sleep disturbances, autonomic dysfunction, and neuronal and glial loss detectable by neuroimaging.[3] There are six isoforms of tau. The predominance and neuroanatomical location of those isoforms vary depending on the tauopathy involved. Alzheimer disease (AD) is a notorious and prevalent tauopathy, characterized by the extracellular accumulation of beta-amyloid [or amyloid β (Aβ)] plaques and spreading of intracellular neurofibrillary tangles (NFTs) in the brain, leading to neuronal damage and death.[4] AD can manifest a variety of symptoms within an individual, most commonly dementia. AD accounts for an estimated 60% to 80% of all dementia cases.[5] According to the World Health Organization, more than 57 million people globally are diagnosed with dementia with cases rising about 10 million annually (World Health Organization; Dementia, *https://www.who.int/health-topics/dementia#tab=tab_2*, last accessed December 1, 2024). In 2019, approximately 5.8 million Americans had AD, a number expected to double by 2050, with projected U.S. healthcare costs nearing 1 trillion USD.[6,7] Amidst a growing elderly population (>65 years), dementia has become a major cause of dependency and socio-economic burden.

The greatest pathological hallmarks of AD are the formation and accumulation of Aβ plaques and NFTs within neurological tissue.[4] Although these features are strongly associated with AD, their exact role in causing neurodegeneration and subsequent cognitive decline, behavioral changes, psychological disorder(s), and whole-body disturbances (eg, alteration of mobility, loss of appetite) remains unclear. Aβ plaques result from the misfolding of the Aβ fragments of 39 to 43 amino acids. The fragmented Aβ peptides are produced from amyloid precursor protein (APP), a transmembrane glycoprotein expressed in neuronal

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synapses involved in neuronal transmission and homeostasis. APP is cleaved by beta- and gamma-secretases to produce Aβ fragments, whose fibrillar form is hypothesized to be the primary component of the Aβ plaques.[8,9] Aβ monomers can aggregate into insoluble amyloid fibrils, which may contribute to the assembly of Aβ plaques.[8,10] These plaques collect extracellularly between neuronal cells, disrupting healthy neurological cell functions such as cellular communication, metabolism, and repair (National Institute on Aging; *https://www.nia.nih.gov/health/alzheimers-causes-and-risk-factors/what-happens-brain-alzheimers-disease*, last accessed December 1, 2024). NFTs are another hallmark of AD, resulting from microtubule-associated tau protein, which normally regulates microtubule activity in the neurons.[11] Tau, along with other microtubule-associated proteins, is tightly regulated by post-translational modifications (PTMs).[12] The most predominant PTM of tau in AD is phosphorylation.[12] The hyperphosphorylation of tau accelerates the aggregation in comparison with the non-phosphorylated form of tau.[13] The aggregation of tau results in the formation of oligomers (initial building blocks for the formation of fibrils), beta-sheet threads, and subsequent formation of insoluble NFTs within the neuron. NFTs interfere with the neuron structure and function such as signal transmission. Upon the formation of fibrillar tau, it can spread to other regions of the brain in a prion-like manner.[12] Amyloid plaques and NFTs ultimately cause neuronal death and brain atrophy (National Institute on Aging; *https://www.nia.nih.gov/health/alzheimers-causes-and-risk-factors/what-happens-brain-alzheimers-disease*, last accessed December 1, 2024). However, the spatiotemporal accumulation of NFTs correlates better with the loss of cognition.[14]

Despite extensive research, a cure for AD has not been found, largely due to its multifactorial nature and incompletely understood pathology. Factors such as aging, genetics, traumatic brain injury, vascular disease, infection, and environmental elements contribute to the disease.[15–17] Because of the unknown underlying cause of AD, most medications for the disease are designed to manage symptoms rather than alter disease progression. Currently, three classes of AD drugs are available—acetylcholinesterase inhibitors, N-methyl-D-aspartate (NMDA) receptor antagonists, and anti-Aβ monoclonal antibodies (Figure 1). Acetylcholinesterase inhibitors, including donepezil (Aricept; Eisai, Woodcliff Lake, NJ), rivastigmine (Exelon; Novartis Pharmaceuticals, Bern, Switzerland), and galantamine (Razadyne; Ortho-McNeil Neurologics, Raritan, NJ), prevent the breakdown of acetylcholine, thereby enhancing cholinergic transmission.[18,19] NMDA receptor antagonists, such as memantine (Namenda; AbbVie, North Chicago, IL), block excessive glutamatergic neurotransmission to reduce excitotoxicity, which is linked to neuronal cell death and dysfunction.[20] Although these two traditional therapies provide symptomatic relief, they do not reverse, cure, or halt disease progression.

Since 2021, monoclonal antibodies that reduce Aβ accumulation in the brain, specifically aducanumab[21]

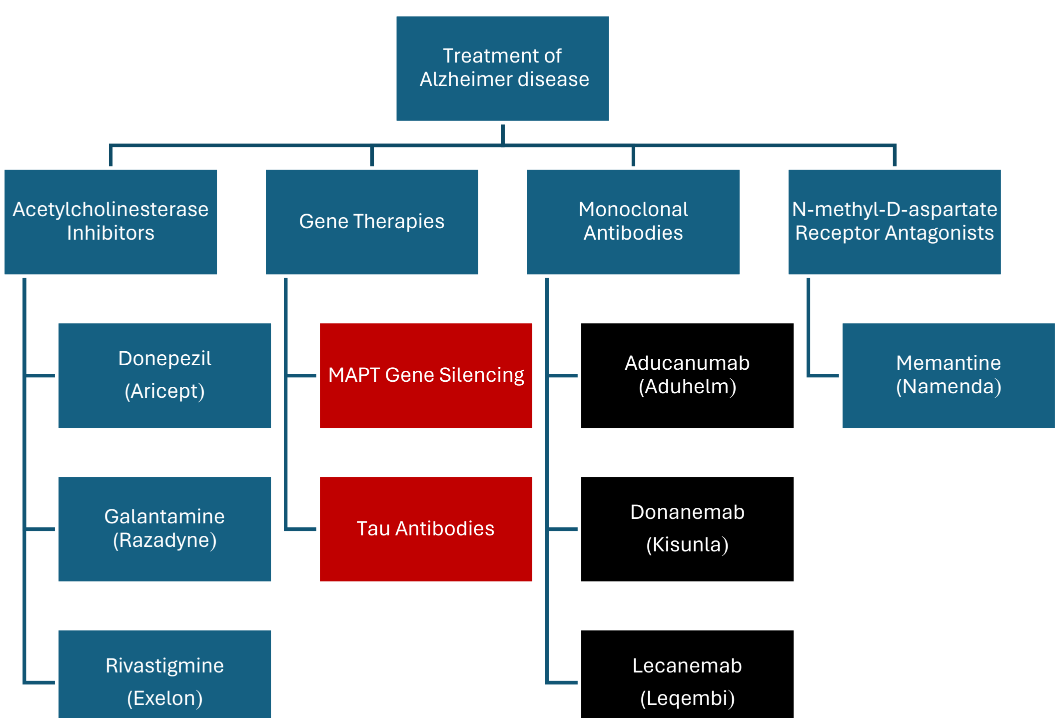

Figure 1 Current and emerging therapeutic approaches in Alzheimer disease. Treatments in **blue boxes** represent broad symptomatic therapies, those within the **red boxes** target tau pathology, and those within the **black boxes** target amyloid-beta.

(Aduhelm; Biogen, Cambridge, MA; withdrawn from the market in 2023), donanemab[22] (Kisunla; Eli Lilly), and lecanemab[21] (Leqembi; Eisai), have received Food and Drug Administration (FDA) approval for the treatment of AD. However, their clinical benefit remains controversial, as safety concerns persist, particularly regarding amyloid-related imaging abnormalities, a common side effect involving brain swelling or hemorrhage.[23,24] Additionally, a sparse number of clinical trials for AD treatments have shown high failure rates for symptomatic agents as well as disease-modifying small molecules and immunotherapies, with 99.6% failing between 2002 and 2012.[25] Therefore, there is an urgent need for the discovery and development of novel therapies that can reduce the progression of neurodegeneration in AD patients or provide a complete cure.

This review aims to summarize novel research findings on promising therapeutic targets related to the pathological features of AD, including tau-related protein targets, post-translational modifications to tau, gene therapies, and novel antibodies that could support effective drug development for this complex disease (Figure 2, Table 1). It is important to note that the microvascular pathology of AD is not discussed in this review.

Promising Therapeutic Strategies

To address the growing concern over the ever-increasing AD cases due to longer lifespans, drug discovery has focused on disease-modifying therapies aimed at slowing disease progression or delaying onset. However, the multifactorial nature of AD means that potential therapeutic targets often overlap with essential biochemical processes, complicating the development of effective disease-modifying therapies for AD. Since 2003, only 2% of AD drug candidates have successfully advanced through phase 2 and 3 clinical trials.[26] Many of these drug candidates in development have faced challenges in clinical trials due to insufficient target engagement or unacceptable side effects, as seen with beta-site APP cleaving enzyme 1 (BACE1) inhibitors and gamma-secretase inhibitors.[27,28] Another limitation is the poor translation of early successes from animal models to humans. This difference may occur from the greater size and complexity of amyloid plaques in human tissue compared with those in mice, difficulties in modeling tau pathology, and the lack of neurodegeneration in many animal models.[29] Additionally, the heterogeneity and individual variability in the progression and development of AD pose another challenge in developing disease-modifying therapies.[30] Given these roadblocks, there have been earnest efforts to push for the discovery of unique and promising therapeutic targets for AD.

With the evolution of molecular biology techniques, a rising number of relationships between target proteins and AD are being identified. These novel targets provide the scientific community with a broader perspective on the multifactorial complexity of AD. Although researchers have yet to fully understand AD's mechanisms, the growing emphasis on target-based drug discovery to modulate the activities of one or more biochemical functions contributing to AD has expanded scientific knowledge of drug

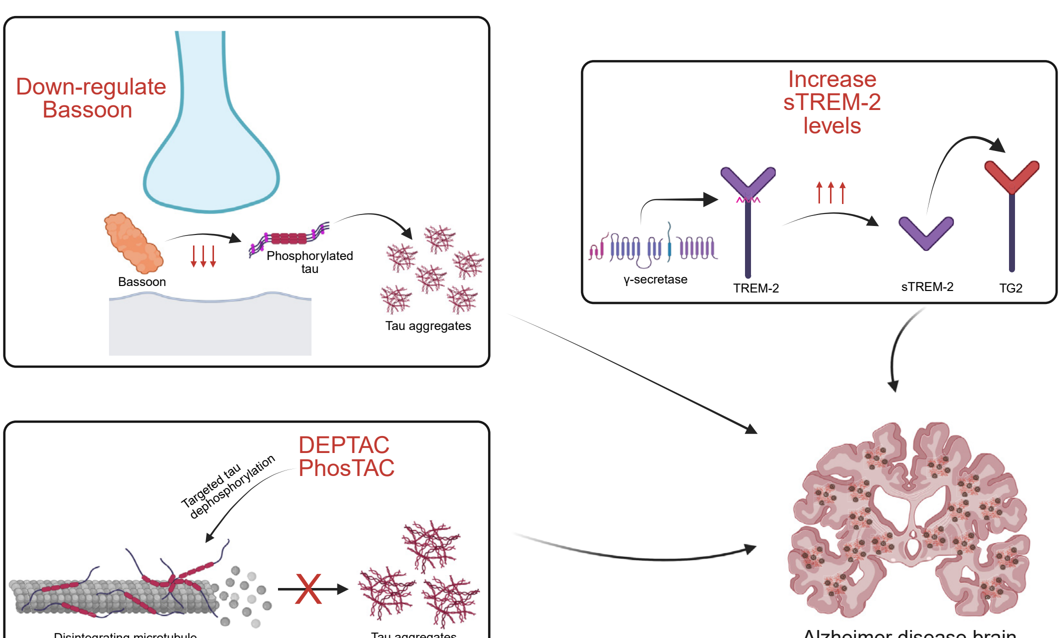

Figure 2 Proposed mechanisms for therapeutic intervention in Alzheimer disease. Down-regulation of the Bassoon protein reduces its interaction with high-molecular-weight tau aggregates, resulting in destabilization of tau seeds and reduced tau pathology. Elevated levels of soluble TREM-2 (sTREM-2) enhances the binding to transgelin-2 (TG2), thereby decreasing tau hyperphosphorylation. The application of dephosphorylation targeting chimera (DEPTAC) and phosphorylation targeting chimera (PhosTAC) promotes tau dephosphorylation and reduces tau aggregation.

Table 1 Mechanisms of Action and Therapeutic Potential of Alzheimer Disease Targets

Topic	Mechanism of action	Therapeutic strategy and benefits
Bassoon	Interacts with high-molecular-weight tau aggregates, stabilizes tau seeds, and promotes tau propagation and neurotoxicity	Down-regulate Bassoon: Destabilize tau seeds ↓ Neurotoxicity ↑ Synaptic integrity ↓ Tau and NFT pathology
TREM2	Binds to TG2, inhibits RhoA/ROCK-GSK3β, and reduces tau phosphorylation, modulates microglial function	Soluble TREM2+ derivatives: ↓ Cognitive decline ↑ Synaptic function ↓ Tau hyperphosphorylation
DEPTAC	Recruits PP2A to dephosphorylate tau, reducing tau phosphorylation and aggregation	DEPTAC: ↑ Cognitive function ↓ NFTs ↓ Tau phosphorylation
PhosTAC	Recruits PP2A to dephosphorylate tau, enabling targeted degradation of phosphorylated tau	PhosTAC: ↑ Cognitive function ↓ NFTs ↓ Tau phosphorylation
OGA inhibitors	Increase O-GlcNAcylation of tau, stabilizing tau and reducing its aggregation and phosphorylation	OGA inhibitors: Stabilize tau ↓ Tau-related neurodegeneration

DEPTAC, dephosphorylation targeting chimera; NFT, neurofibrillary tangle; PhosTAC, phosphorylation targeting chimera.

formulation for this complex disease. However, several studies have shown weak correlations between *in vitro* and *in vivo* drug effects for target-specific drug discovery.[31] It is hypothesized that this is because AD, in addition to biochemical mechanisms, has multiple interplaying genetic and environmental factors unique to each individual.[32] Therefore, efforts to identify novel targets and expand the knowledge of complex relationships between protein targets and AD will allow for a clearer understanding of the disease and create a step forward in developing effective multi-target disease-modifying therapies. This section of the review discusses recent notable discoveries in relationships between AD and various protein targets that provide a better understanding of the mechanisms of AD pathogenesis.

Bassoon Protein Interacts with High-Molecular-Weight Tau Complexes to Exacerbate Tau Pathology

Current understandings suggest tau proteins interact with other proteins and complexes that support microtubule stability and various biological functions. However, additional research is necessary to compare the interactomes of seeding-component tau to monomeric tau and how tau—seed interactors influence tau propagation. Martinez et al[33] characterized tau species involved in propagation and identified proteins directly interacting with tau seeds, notably the scaffolding protein Bassoon (BSN). A small fraction of tau forms a high-molecular-weight complex with strong seeding activity, interacting with proteins like BSN, which do not interact with tau monomers.[33] Targeting high-molecular-weight pathogenic tau aggregates, instead of nonpathogenic tau monomers, may provide benefits. Focusing on the BSN-—high-molecular-weight tau complex can more effectively neutralize toxic tau aggregates, potentially reducing neurotoxicity and disease progression while minimizing side effects associated with reduced tau monomers such as microtubule instability, synaptic dysfunction, axonal transport impairment, disruption of signal transduction, and altered stress response.[34–37] The co-localization of BSN protein with tau in postmortem brain samples from AD and progressive supranuclear palsy patients, where the tau seed interacts with BSN at the presynaptic compartments in early pathology and co-deposits with BSN in the cytoplasm as the disease progresses.[33] Martinez et al[33] thoughtfully investigated the specific effects of the BSN protein in terms of tau spreading and pathophysiological effects. Additionally, Martinez et al[33] explored the disease-altering effects of adjusting BSN levels in the brain to point toward its therapeutic potential.

Overexpression of BSN with P301S tau in HEK293T cells increased tau seeding activity and the subsequent accumulation of misfolded, aggregated tau. BSN directly interacts with misfolded or aggregated tau through its C- or N-terminal regions in a conformation-dependent manner and does not interact with non-aggregating tau.[33] This interaction is partly due to BSN's large molecular weight, disorganization, and low hydrophobicity, subjecting it to abnormal interactions with tau aggregates. In *Drosophila melanogaster*, BSN overexpression did not cause degeneration alone but worsened tau-induced eye degeneration. Both BSN forms increased misfolded tau accumulation and tau-seeding activity, with mutant BSN having a stronger effect.[33] Overall, BSN promotes pathological tau aggregation, contributing to tauopathies and neurodegeneration, which led Martinez et al[33] to further study the potential beneficial effects of down-regulating BSN.

To investigate whether BSN is essential for misfolded tau spreading in the brain, an adeno-associated virus (AAV) was used to transmit mRNA expressing both GFP and htauP301L. Neonatal mice were injected with AAV containing either shRNA against mouse BSN (shBSN) or control shRNA (scramble), and it was found that shBSN reduced BSN expression by approximately 60% without negatively affecting presynaptic integrity or producing abnormalities. The mice received hippocampal injections of AAV-GFP-(P2A)-hTauP301L 3 months post-shRNA, resulting in the BSN knockdown mice showing reduced tau spreading compared with control mice.[33] These results were confirmed in PS19 mice, where similar procedures showed that BSN down-regulation decreased NFTs, misfolded tau, microgliosis, and astrogliosis, which resulted in increasing synaptic integrity.[33] Reduced BSN did not have a direct effect on total tau, but drastically reduced tau-seeding activity in high-molecular-weight tau seeds, making tau aggregates more prone to degradation.

Overall, BSN plays a crucial role in stabilizing misfolded tau aggregate, enhancing their pathogenic capacity and contributing to the development of tauopathies such as AD. However, the systematic down-regulation of BSN proved promising in reducing tau pathology and improving synaptic integrity of the PS19 mouse model. Concerning tauopathy-related clinical signs, Martinez et al[33] found that BSN down-regulation improved motor impairment and rescued decreased body temperature in male and female PS19 mice. However, BSN down-regulation may only partially reduce neurodegeneration in mice, as no differences in hippocampal size were observed between PS19-shBSN mice and WT-scramble or WT-shBSN mice. Finally, BSN down-regulation of PS19 mice in the early stages of tau aggregation reduced phosphorylated tau aggregates and provided a lasting effect on tau burden as pathogenesis progressed.[33]

Martinez et al[33] provided thorough research on the BSN protein, identifying it as a tau−seed interactor with a significant role in tau−seed stability and spreading in the brain, promoting tau pathogenesis. In this study, BSN was shown to be a promising protein target for neurodegenerative tauopathies such as AD, as its down-regulation in animal models significantly reduced tau spreading and the related neuropathological lesions. It also improved physiological mechanisms that are typically impaired by pathogenic tau, such as synaptic integrity, electrophysiology, behaviors, and brain mass. In addition, BSN down-regulation produced no abnormalities in mice models, suggesting that a 50% reduction in BSN is well-tolerated.[33] However, BSN's exact role in tau pathogenesis remains unclear. Although Martinez et al[33] provide significant evidence that BSN promotes pathogenesis by stabilizing tau aggregates, BSN also plays a role in presynaptic autophagy through the protein ATG5,[38] a vital process for clearing toxic materials and maintaining presynaptic function, as well as regulating presynaptic ubiquitination.[39] The interaction of BSN with ATG5 led to the inhibition of the induction of autophagy in cell-based assays.[38] Tai et al[40] reported that the accumulation of hyperphosphorylated tau oligomers in AD relates to the dysfunction of the ubiquitin-proteasome system, providing additional evidence of the importance of studying the role of BSN in synaptic health. Thus, further exploration of BSN's functions is needed before developing a BSN-targeted treatment for tauopathies.

Soluble TREM2 Provides a Potentially Beneficial Effect in AD by Reducing tau Phosphorylation at Vital Sites by Binding to Transgelin-2

Triggering receptor expressed on myeloid cells 2 (TREM2) consists of a transmembrane receptor found in myeloid cells such as microglia. When cleaved by α-secretase (ADAM10, ADAM17), it releases soluble TREM2 (sTREM2).[41] sTREM2 is elevated in the cerebrospinal fluid (CSF) of AD patients, especially in late-presymptomatic and early symptomatic stages.[42] Higher levels of sTREM2 correlate to diminishing cognitive decline, suggesting potential benefits in AD.[43] Genetic studies showed that rare variants of TREM2 can increase the risk of late-onset AD by 2- to 3-fold, with AD risk variants such as R47H and R62H impairing TREM2 ligand binding and altering microglial function.[44,45] Additionally, loss-of-function variants in TREM2, including homozygous variants linked to early-onset dementia and heterozygous variants associated with later-onset AD, further elevate disease risk and contribute to increased tau accumulation and brain atrophy in the presence of amyloid deposits and related changes.[46−48] Conversely, the potential gain-of-function T96K mutation in TREM2 may reduce AD risk through increased cellular binding.[49] However, findings remain conflicting, as T96K carriers have greater memory and language impairments, yet show lower levels of CSF-Aβ42, phosphorylated tau-181, and total tau compared with noncarriers.[50] TREM2 also controls the innate immune system, activating astrocytes and microglia in response to Aβ.[51] However, adjusting TREM2 levels in the brain yield contradicting results. Some studies show that silencing TREM2 exacerbates tau-related neuropathology,[52,53] whereas overexpression reduces the downstream neuropathological changes by tempering down the microglial inflammation.[54] Other studies using transgenic mouse models suggest that TREM2 deletion (or reduction) has no direct impact on tau phosphorylation or insoluble tau levels,[55,56] though chronic TREM2 activation increases Aβ-induced tau seeding and spreading of misfolded tau.[57] Considering these perplexing findings, further research is needed to define TREM2's role in tau phosphorylation and misfolding as well as neuroinflammation in AD, with potential implications for disease-modifying treatments.

Zhang et al[58] aimed to investigate sTREM2's effects on tau phosphorylation in neurons to further understand its phosphorylation mechanisms. They found that sTREM2 reduced tau phosphorylation at specific sites such as S202, S396, T181, and S404, with the most dramatic dephosphorylation at S202 and S396. Notably, these sites are modified by the glycogen synthase kinase-3β (GSK3β), an enzyme linked to AD due to its tau-phosphorylating function and Aβ production. GSK3β levels in the brain increase with age[59] and its activity is overactive in AD brains,[60] making it a major player in the pathophysiology of AD. Zhang et al observed that GSK3β phosphorylation at Y216 significantly decreased in a concentration-dependent manner in the presence of sTREM2.[58] Moreover, purified sTREM2 decreased tau phosphorylation and GSK3β activation that were induced by the presence of Aβ oligomers.[58] These results suggest that sTREM2's mechanism of attenuating tau hyperphosphorylation is linked to its role related to GSK3β.

Zhang et al[58] identified transgelin-2 (TG2) as a surface-cell receptor interacting with sTREM2 in the neural membrane fraction. TG2 is involved in the RhoA/ROCK pathway, where its activation increases RhoA phosphorylation at S188, hence inactivating the pathway.[61] The RhoA/ROCK pathway is known to activate GSK3β.[61] The microglia-derived sTREM2 binds to TG2, induces RhoA phosphorylation at S188, and deactivates the RhoA-ROCK pathway, increasing tau phosphorylation.[61] sTREM2 binding to TG2 inhibits the phosphorylation and activation of RhoA, thereby activating the RhoA/ROCK pathway and inactivating GSK3β.[58] In AD patients, TG2 and phosphorylated RhoA levels were decreased while GSK3β phosphorylation level of a specific tyrosine, that is, Y216, was increased.[58] Thus, the TG2-RhoA-ROCK-GSK3B pathway is perturbated in AD, suggesting that sTREM2 or its derivatives could serve as promising therapeutics for AD. To mimic the effect of sTREM2, Zhang et al[58] designed the sTREM2 (77 to 105) active peptide and fragmented it further into peptides 1 to 4. Peptide 1 (amino acids 77 to 89) was found to inhibit tau phosphorylation the most and inactivate the RhoA-ROCK-GSK3β pathway. It also attenuated GSK3β activation that was induced by Aβ oligomers and, subsequently, reduced tau phosphorylation.[58] Peptide 1 effectively bound to neurons via TG2, inhibiting tau phosphorylation. Tested on P301S mice, peptide 1 was brain-permeable and detected in multiple brain regions up to seven days post-injection. It significantly reduced tau phosphorylation and inhibited the activation of both GSK3β and RhoA in the hippocampus. The treatment resulted in improved synaptic function, learning, and memory in P301S mice in comparison with the group control.[58]

Zhang et al[58] discovered a novel function of sTREM2 and TG2 in halting the progression of tau induced neuropathology and identified the minimal active sequence (amino acids 77 to 89) that mimics this protective effect. This study provides a basis for targeting this interaction therapeutically in AD, potentially through up-regulation of TG2 or sTREM2. However, contradictory studies regarding sTREM2 require further attention, as the mechanisms of TREM2 in the formation of NFTs and induced neuropathology remain unclear. Multiple studies conclude that TREM2 plays a vital role in tau causative roles in induced neuropathology, albeit with varying results. As Zhang et al[58] mentioned, this variability could stem from different mice models or methods of TREM2 manipulation. Higher levels of TREM2 are linked to diminished cognitive decline, Aβ deposition, and NFTs related neurodegeneration. However, the molecular mechanisms of sTREM2 on the induction of tau phosphorylation is ambiguous. sTREM2 potently inhibits tau phosphorylation at S202, S396, S404, T181, but not at T231. This highlights the need for further investigation into tau phosphorylation at T231, as it appears unrelated to sTREM2.

Additionally, Zhang et al[58] found that the TG2-sTREM2 interaction is active in both basal and AD conditions, posing the question of how this interaction is altered during AD, reducing its ability to inhibit hyperphosphorylation. Future findings will be key to clarifying the exact role and mechanism of TREM2 in tau phosphorylation and its potential therapeutic targeting. Cap et al showed that Aβ activates the RhoA-ROCK-GSK3B signaling pathway, leading to tau phosphorylation,[61] whereas Zhang et al[58] demonstrated that sTREM2 binds to TG2 and inhibits this pathway to reduce tau phosphorylation. Therefore, the primary activation of Aβ on the RhoA-ROCK-GSK3β pathway negates sTREM2's beneficial effects on tau pathology. This prompts further investigation into how Aβ activates this pathway and how to optimize the mechanisms of sTREM2 to effectively reduce tau phosphorylation in AD. Zhang et al[58] also found that microglia secrete sTREM2 to protect neurons against developing tau pathology in earlier stages of AD. However, the communicatory mechanisms between microglia and neurons and their contributions to tau pathology are poorly understood. This prompts yet more investigations into the role of microglia in AD and how they can be optimized as a therapeutic intervention.

Targeting the Hyperphosphorylation of tau through Phosphorylation-Targeting Chimera Molecules Aids in Reducing Phosphorylated Tau by Targeting PP2A

Tau is heavily regulated by PTMs, with phosphorylation being the most studied due to its connection to AD. Abnormal aggregation of tau into higher-order structures called oligomers is a hallmark of AD.[62] In AD, tau is hyperphosphorylated due to mutant genes or environmental factors, leading to insoluble aggregates and cognitive decline.[12] The exact mechanism by which hyperphosphorylation contributes to tauopathies like AD is not coherently known, but several hypotheses exist. One prominent theory suggests that phosphorylation at Thr23 causes tau to undergo an isomerization change from a *cis* to a *trans* conformation, reducing its binding affinity to microtubules and leading to their disassembly.[63,64] Hyperphosphorylated tau tends to aggregate and self-polymerize, forming paired helical filaments (PHF) and straight filaments (SF) *in vitro* that develop into neurofibrillary tangles (NFTs).[65] Dephosphorylation of tau resulted in the inhibition of tau self-polymerization, restoring its normal function in stabilizing microtubules.[66] Thus, targeting tau hyperphosphorylation in AD has been given significant attention by researchers in the pursuit of developing mechanisms to ameliorate pathology associated with hyperphosphorylated tau.

To achieve efficient clearance of hyperphosphorylated tau at the post-translational level, Zheng et al[67] developed dephosphorylation targeting chimera (DEPTAC) to promote the binding of tau to PP2A—specifically its regulatory subunit B, which targets substrates. PP2A is a major serine and threonine phosphatase involved in numerous cellular processes, including tau phosphorylation.[68] PP2 dysfunction is linked to tau hyperphosphorylation, and Mondal et al showed that activating PP2A can mitigate the progression of AD.[69] Thus, targeting PP2A has become a focus in developing AD therapeutics. DEPTAC is a small peptide (~4.4 kDa) composed of 38 amino acids with four functional motifs: tau binding, a flexible linker, PP2A-Ba recruitment, and cell penetration. DEPTAC selectively binds tau while recruiting PP2A-Ba and shows high neuron penetrability and low cytotoxicity. It significantly reduced tau phosphorylation in the rat hippocampal neurons overexpressing human tau without significantly altering MAP2, indicating a specificity to tau. DEPTAC-induced tau dephosphorylation occurred independently of total tau reduction, even with proteolysis facilitating this process.[67] The peptide's effects on phosphorylated and total tau clearance suggest potential benefits for treating tau pathologies in AD.

DEPTAC's efficacy in dephosphorylating tau was successful in cultured neurons and tested in 9-month-old 3xTg AD mice, receiving DEPTAC repeatedly over a month.[67] After the treatment, DEPTAC significantly reduced phosphorylated tau and total tau levels in the 3xTg AD mice hippocampus. Its efficiency was further assessed in Tau368 mice that overexpressed the human tau N-terminal 1 to 368 fragment (hTau-N368) to avoid interferences from APP and PSEN1 mutations.[67] Following hTau-N368 induction with doxycycline, DEPTAC treatment reduced the hippocampal phosphorylated tau and total tau levels 12 to 72 hours post-injection over a month period.[67] Chronic DEPTAC treatment facilitated tau dephosphorylation and degradation over three months of continuous hTau-N368 expression.[67] In Tau368 mice without doxycycline, DEPTAC significantly down-regulated phosphorylation of endogenous mouse tau at specific sites such as Thr205, Ser396, and Ser404. However, DEPTAC had limited effects on the phosphorylation of specific amino acid(s) (such as Ser199) and the expression level of total tau.[67]

DEPTAC was evaluated for its effect on tau-related hippocampal biochemical changes in hTau mice. After a month of DEPTAC administration following the cessation of doxycycline-induced hTau-N368 expression, both hippocampal phosphorylated tau and total tau levels were reduced. Notably, DEPTAC lowered tau aggregates in the dentate gyrus.[67] Since abnormal hyperphosphorylated tau accumulation impedes neurite differentiation and synapse formation,[70] DEPTAC treatment impact on the structural arrangement of neurons was examined. DEPTAC significantly increased dendritic spine densities and mossy fiber puncta areas in the dentate gyrus. These specific changes in densities and surface areas indicate improved neurite plasticity. In addition, DEPTAC reversed microtubule disassembly caused by hTau-N368 overexpression, improving microtubule assembly and stability.[67] In Tau368 mice, DEPTAC improved the hippocampus-dependent cognitive impairments from doxycycline-induced hTau-N368 overexpression, such as object recognition and spatial learning.[67]

DEPTAC is a promising treatment method that targets phosphorylated tau at the post-translational level, suggesting it could be used in early stages of AD before hyperphosphorylated tau aggregates disrupt neurological function. While the accumulation of hyperphosphorylated tau is not the sole cause of AD, it mediates synaptic toxicity from Aβ. Therefore, facilitating tau dephosphorylation in AD brains may help mitigate disease progression. Zheng et al studied this novel therapy in multiple AD models, with potential for broader application in other tauopathies.[67] DEPTAC effectively degrades and dephosphorylates tau, which is beneficial in the AD state. However, tau phosphorylation is vital for microtubule function, so DEPTAC's therapeutic mechanisms must be researched to ensure benefits outweigh possible side effects. Zheng et al confirmed no significant differences in treated mice compared to controlled until six months old.[67] DEPTAC's degree of tau reduction is less than other knockdown mechanisms and does not significantly alter neuronal structure or function, suggesting it may be a safer option. However, translating this technology into an FDA-approved drug is particularly challenging, as DEPTAC has been shown to be effective only at concentrations above 200 μmol/L and within three days post-administration.[67] Its short duration of action limits its drug development viability. Peptides are subject to proteolysis, likely contributing to DEPTAC's with limited duration of efficacy. Improving the stability of these chimera drugs, extending their efficacy, enhancing blood-brain-barrier penetration, changing formulation, exploring intranasal route of administration, and achieving effectiveness at lower concentrations would strengthen their potential for AD treatment.

A recently developed method of targeted tau dephosphorylation offers potential therapeutic use for AD, using a technique comparable to DEPTAC. A novel approach that avoids non-selectively activating protein phosphatase activity has been developed, which can cause unwanted side effects due to their roles in numerous vital functions, such as smooth muscle contraction.[71,72] Phosphorylation targeting chimera (PhosTAC) is a method that uses heterobifunctional molecules to promote the complex formation between a phosphatase and a targeted phosphorylated protein, allowing for focused targeting compared to non-discriminatory phosphorylation methods.[72] PhosTAC adapts the research group's previous PROTAC technology, which utilizes bifunctional small molecules to induce protein ubiquitination and degradation, offering advantages over traditional inhibitors, including catalytic mechanisms, high selectivity, and prolonged effects.[73–75] Previous successes with PROTAC techniques targeting PDCD4 and FOXO3a provide evidence for applying PhosTAC methods to tau dephosphorylation.

A mammalian cell system was engineered to express full-length 2N4R and 2N3R tau variants, along with P301L and S320F variants, which accelerate the formation of NFTs.[76] The phosphatase used in this study was PP2A, the major tau phosphatase in humans.[68] PhosTAC is composed of an FKBP12(F36V) engineered ligand and a HaloTag ligand that recruits PP2A to dephosphorylate HaloTag-fused tau. By using genetically encoded tags and chemical recruiting elements, PP2A formed ternary complexes that induced rapid tau dephosphorylation. Notably, tau dephosphorylation was sustained even 24 to 48 hours after PhosTAC post-treatment. Additionally, forced tau dephosphorylation enhanced the degradation of tau.[72] PhosTAC induces dephosphorylation and degradation of tau. PhosTAC exploits a promising strategy for the development of new AD therapeutics. The advantage over traditional kinase and protease inhibitors from this PhosTAC strategy is the direct targeting of tau. In addition to reducing tau phosphorylation, PhosTAC has potential in exerting broader therapeutic impacts indirectly. By reducing tau hyperphosphorylation, biological processes in the brain such as microtubule stabilization, synaptic function, and microtubule transport could be ameliorated. This suggests that PhosTAC could have significant potential in reducing the symptoms associated with AD by targeting tau phosphorylation.

In conclusion, the studies discussed in this section employ chimeric molecules DEPTAC and PhosTAC as promising strategies for targeting tau phosphorylation in AD. Both approaches address tau hyperphosphorylation, which contributes to neurofibrillary tangles and neurotoxicity in AD. DEPTAC promotes tau dephosphorylation by facilitating its binding to PP2, with Zheng et al demonstrating its efficacy in reducing phosphorylated tau levels in cultured neurons and mouse animal models.[67] Its high specificity for tau and ability to penetrate neurons with low cytotoxicity are significant advantages over other methods. However, its short duration of action and high effective concentration must be addressed to develop a viable long-term treatment option for AD. PhosTAC recruits phosphatases to phosphorylated tau, inducing tau dephosphorylation and degradation. This highly specific technique offers potentially fewer side effects compared to broad-spectrum PP2A activators. While PhosTAC's ability to dephosphorylate and degrade tau highlights its therapeutic potential, further investigation with diverse models and long-term studies is needed to prove its safety and efficacy. The future of phosphorylation-targeting chimeras in AD appears promising, but continued research is essential to enhance stability, duration of action, and delivery systems for effective treatments.

OGA Inhibitors

While hyperphosphorylation is the most notable PTM, O-GlcNAcylation is being explored as a promising target for AD therapy. During O-GlcNAcylation, a N-acetylglucosamine (GlcNAc) group is added to the hydroxyl group of a serine or threonine residue of nucleic and cytoplasmic proteins like tau, regulating several cellular processes such as signal transduction, transcription, and stress response.[77] Abnormal O-GlcNAcylation can disrupt these processes and contribute to various diseases, including neurodegeneration. Two enzymes regulate this PTM. The O-GlcNAc Transferase (OGT) is responsible for the addition of the GlcNAc, while O-GlcNAcase (OGA) cleaves it.[77] Recent data suggest that O-GlcNAcylation rates decrease in the aged healthy brain and AD brains, likely due to metabolic abnormalities and mitochondrial dysfunction.[77] Specifically, Liu et al[78] reported that O-GlcNAcylation levels in AD brains were 22% lower than the levels found in healthy brains. Interestingly, the hyperphosphorylated tau exhibited four times less O-GlcNAc groups in comparison with non-hyperphosphorylated tau.[78] Therefore, the inhibition of OGA may constitute an attractive strategy relevant to AD pharmacotherapy due to the reduced O-GlcNAcylation found in AD.

Two compounds that serve as OGA inhibitors have been explored in pre-clinical rodent models. Developed in 2008, Thiamet G is a selective and potent OGA inhibitor, leading to reduced tau phosphorylation at two specific sites, that is, Thr231 and Ser396.[79] Thiamet G decreased cellular internalization of α-synuclein fibrils, alleviating degeneration in dopaminergic neurons by reducing α-synuclein's aggregation, enhancing the neuronal stress response, and facilitating the dopaminergic neurotransmission.[80] A more recent inhibitor, MK-8719, attenuates tau aggregates and hyperphosphorylated tau, reduces total tau in the CSF, and minimizes brain atrophy.[81] Both Thiamet G and MK-8719 are designed to mimic OGA's natural substrate. Although they exhibit high specificity and reduced off-target effects, their effectiveness *in vivo* is limited by poor brain penetration and high effective concentrations. Therefore, exploration with more biologically effective mechanisms to target OGA has been initiated within the field.

Permanne et al[82] have developed a novel selective, substrate-competitive OGA inhibitor with an excellent brain permeability and target selectivity profile named ASN90. Its chemical composition is different from Thiamet G and MK-8719 as it is not carbohydrate-based. ASN90 significantly increased OGA protein levels without significant alteration in OGT protein expression. Several phase 1 clinical trials have been completed with ASN90 in healthy human subjects, showing promising results. The trials indicated that ASN90 was safe and well-tolerated at the dose administered within the treatment scheme evaluated. There were dose-limiting toxicities or serious adverse events reported.[82] These findings support further clinical development in Phase 2 clinical trials, where ASN90 will be tested in patients with tauopathies to determine its effectiveness.[83] ASN90 shows great promise as a potential AD therapy.

Gene Therapies

Emerging treatments for AD, particularly anti-amyloid monoclonal antibodies derived from cloning single white blood cells, have shown promise but remain expensive and require frequent administration. A novel approach uses tau antibodies to target the abnormal accumulation and hyperphosphorylation of tau protein, a primary marker of AD.[84] Chang et al[85] demonstrated that, despite the challenges posed by the brain's blood-brain barrier, certain protein treatments can cross the blood-brain barrier and reduce amyloid peptide accumulation in the hippocampus of mice. Consequently, conventional tau antibodies may inhibit the prion-like spread of extracellular tau, whereas strategies for targeting intracellular tau are still in development. Gene therapies present another potential solution. Wongsodirdjo et al[84] successfully targeted intracellular tau using an antibody transcribed from synthetic, *in vitro* transcribed mRNA. This engineered mRNA encodes tau antibodies that require cytosolic uptake, offering faster and more cost-effective production than viral methods, making it a safer option for intrabody therapy in neurodegenerative diseases. The FDA's approval of gene therapies for spinal muscular atrophy further supports the potential of this approach for treating neurodegenerative conditions.[86] Wongsodirdjo et al[84] also found that human neuroblastoma SH-SY5Y cells successfully transfected with mRNA encoding the tau-specific antibody RNJI produced RNJI in IgG format, indicating that although mRNA does not affect transfection, it influences the localization of the translated protein.

An alternative approach to gene therapy involves reducing the expression of the tau gene, microtubule-associated protein tau (MAPT). In AD patients, mutations in MAPT increase tau aggregation, leading to neurotoxicity. Previous studies have shown that mice lacking MAPT are protected from neuronal damage and memory deficits.[87,88] Suppressing the MAPT gene in these mice reduces the harmful effects of tau accumulation.[88] This suggests that suppressing tau could be a promising therapeutic approach for dementia, though it requires intracellular treatments that effectively reach tau in the brain.

Antisense oligonucleotides and anti-tau antibodies have been shown to be effective, though chronic administration is needed to show progress in reducing tau. Additionally, Wegmann et al[89] proposed two methods using AAV to deliver treatments: intracranial injections into the hippocampus and intravenous delivery that can cross the blood-brain barrier. Zinc finger protein transcription factors (ZFPs-TFs) were used to silence MAPT and reduce endogenous tau expression. A single administration of AAV via either method resulted in dose-dependent suppression of tau mRNA and protein isoforms by 50% to 80% for up to 11 months. This tau repression also led to a reduction of Aβ plaques associated with neuritic dystrophy.[89]

In a phase 1b randomized, double-blind, placebo-controlled trial, $MAPT_{RX}$—a tau-targeting antisense oligonucleotide that binds to intron 9 of the MAPT pre-mRNA and inhibits tau protein translation through MAPT mRNA degradation—was evaluated.[90] A total of 46 patients were randomized to different treatment groups: placebo ($n = 12$), $MAPT_{RX}$ 10 mg monthly ($n = 6$), $MAPT_{RX}$ 30 mg monthly ($n = 6$), $MAPT_{RX}$ 60 mg monthly ($n = 9$), or $MAPT_{RX}$ 115 mg quarterly ($n = 13$). The trial found that higher doses of $MAPT_{RX}$ led to a dose-dependent and sustained reduction in CSF total tau levels. In the 60 mg and 115 mg MAPTRX groups, there was a >50% reduction from baseline in CSF total tau concentration at 24 weeks post-treatment. The most common side effect was post-lumbar puncture headache, but no serious adverse effects were reported. This clinical trial marks a promising starting point for treatments that suppress MAPT expression. It demonstrates the potential of MAPT-targeting therapies in reducing CSF total tau concentrations with a good safety profile. However, larger and longer trials will be necessary to confirm these findings and further evaluate the treatment's long-term effects.

These studies emphasize significant progress in addressing the complexities of AD treatment. Both the RNJI antibody and MAPT suppression approaches offer advantages and drawbacks, requiring further development before clinical application. The first study avoids the risk of patients developing resistance to the AAV method through neutralizing antibodies, whereas the second approach targets tau at the transcriptional level. Notably, RNJI was tested *in vitro* using human neuroblastoma SH-SY5Y cells, whereas ZFP-TFs were tested *in vivo* on adult wild-type mice. Despite these differences, both studies represent important strides in advancing gene therapies for neurodegenerative diseases.

Tau-Targeting Monoclonal Antibodies

Recently approved treatments for AD include monoclonal antibodies that specifically target one of its key pathological features, the Aβ plaques. However, immunotherapy has also been explored as a strategy to target tau in search of an effective tauopathy treatment aimed at reducing tau accumulation. Notably, tau-targeting antibodies can cross the blood-brain barrier to reach tau in the brain when administered peripherally.[91] Further, Rodrigues Martins et al[92] developed anti-tau monoclonal antibodies into single-chain variable fragments (scFvs), which can be expressed in the cytoplasm of cells to target intracellular tau. Although tau-specific intrabodies are effective for intracellular tau, scFvs demonstrate greater efficiency in targeting extracellular tau.[93] Additionally, Mukadam et al's[94] research on TRIM21, an antibody receptor and ubiquitin ligase, shows promise in enhancing the effectiveness of antibodies targeting tau (passive immunization) in mouse models. Moving forward, effective AD treatments must advance therapeutic strategies across all levels, from gene expression to the targeting of intracellular and extracellular proteins.

Seeds in the Context of Co-Pathology

AD has been linked to type 2 diabetes due to the protein misfolding involvement in both conditions. In AD, the protein misfolded consist of Aβ and tau, whereas in type 2 diabetes, the amyloid deposits are composed of islet amyloid polypeptide (IAPP). This shared feature has led to AD being referred to as type 3 diabetes or diabetes of the brain due to the similar pathological processes involving insulin resistance in both diseases.[95] In AD, insulin resistance in the brain can impair neurocognitive function and contribute to neurodegeneration. Furthermore, factors such as Aβ toxicity, tau hyperphosphorylation, oxidative stress, and neuroinflammation, which are characteristic of AD, can worsen insulin resistance, amplifying neurodegenerative changes in the brain. Moreover, a pooled analysis of 14 studies involving 2,310,330 individuals and 102,174 dementia cases found that diabetes is associated with a 60% increased risk of developing any form of dementia.[96] The multivariable-adjusted pooled relative risk (RR) for any dementia was 1.68 (95% CI, 1.64-1.71) for women and 1.61 (95% CI, 1.42-1.83) for men.

Aβ and IAPP co-localize in both the brain and pancreas. Aggregated Aβ and hyperphosphorylated tau are present in the Langerhans islets of the pancreases of type 2 diabetes patients.[97] Immunohistochemistry has also shown co-localization of Aβ with IAPP in islet amyloid deposits. Additionally, IAPP has been detected in the temporal lobe gray matter of the brains of patients with diabetes.[98] Therefore, it is evident that the two amyloidogenic proteins, Aβ and IAPP, are in association with each other.

Further investigations into the cross-seeding mechanisms between IAPP and Aβ have provided deeper insights into their interaction. IAPP aggregates can promote the misfolding and aggregation of Aβ *in vitro* through a cross-seeding mechanism, potentially contributing to the progression of more severe AD pathology.[99] Bimolecular fluorescence complementation studies have shown that cross-seeding between Aβ and IAPP leads to heterologous interactions, with a preferential parallel binding orientation that facilitates the formation of amyloid fibrils.[100] Computational analysis further revealed that during cross-seeding, IAPP fibrils catalyze the growth of Aβ fibrils by promoting fibril nucleation and forming parallel, in-register β-sheets at the C-terminus.[101] These findings suggest that a cross-seeding mechanism between IAPP and Aβ likely exists, which could potentially exacerbate the progression of both diseases.

Tau misfolding processes might be influenced by IAPP fibrils or oligomers, which also act as template (seed). IAPP co-localizes with hyperphosphorylated tau in AD brains and promotes its aggregation.[102] These findings suggest that IAPP can cross-seed tau, thereby contributing to tau-mediated neuropathology in AD. *In vivo* injections of IAPP aggregates into tau transgenic mice induce tau phosphorylation, cognitive decline, and synaptic loss.[102] The possible hetero-cross seeding of Aβ and tau by IAPP highlights the importance of identifying molecules capable of inhibiting the seeding and propogation of pathological processes. Targeting seeding mechanisms represents a promising therapeutic approach in the context of AD co-pathology.

Conclusions

Pursuing conventional and emerging targets remains critical in order to diversify and enhance AD pharmacotherapy. Recent advances in understanding AD molecular mechanisms have identified several promising tau-related protein targets for drug discovery. In particular, PTMs of tau in the context of AD will be essential for developing robust therapeutic strategies. Recent studies have demonstrated the potential to finely modulate tau protein expression using novel technologies targeting the MAPT gene. In addition, innovative immunotherapeutic approaches aimed at clearing misfolded tau while minimizing the risk of ARI are urgently needed. This review underscores the importance of pursuing diverse pipelines in molecular biology and drug discovery, exploring both established and novel therapeutic avenues to address this complex multifactorial and chronic neurodegenerative disease.

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Author Contributions

A.D.-T. and N.G.H. wrote the manuscript and designed the figures; J.S.F. edited the manuscript; and all authors read and approved the final version of the manuscript.

Disclosure Statement

None declared.

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